Self-Interacting Dynamics with Exponential Memory: Markovian Embedding, Contractive Memory Fibres, and Metastability Diagnostics

H. Horn

Peine Stederdorf, Germany

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Abstract

This technical note specifies the mathematical structure used by a companion numerical study of self-interacting stochastic dynamics with exponential forgetting. A visible state x_n is coupled to a deposited memory field ρ_n. The visible trajectory is generally non-Markovian, while $z_n = (x_n, \rho_n)$ is a time-homogeneous Markov process. We derive the exact exponential-memory expansion, prove contraction of the affine memory fibre along a common visible path, distinguish one-step invertibility from loss of the ordered history, and state a formal continuum scaling. We also formulate control-aware metastability diagnostics without asserting that the current scalar model has isolated a nonlinear metastable state. The note is intended as a technical appendix or companion reference, not as a standalone novelty claim or numerical phase study.

I. INTRODUCTION

Self-interacting stochastic processes couple a trajectory back to a summary of its past. Classical examples include reinforced processes, Brownian polymers, true self-avoiding or self-repelling motion, and self-interacting diffusions [1–5]. Benaïm, Ledoux, and Raimond use a normalized cumulative occupation measure; for symmetric interactions, Benaïm and Raimond prove convergence toward the critical set of a nonlinear free-energy functional [6, 7]. Herrmann and Roynette study an unnormalized interaction with the full past and establish convergence or boundedness in particular one-dimensional self-attracting cases [8].

Those results do not directly transfer to the model below because its history is exponentially discounted and represented by a deposited spatial field. The closest current comparison is the aging-kernel model of Milišić, Meunier, and Roux, which treats weighted linear interactions with past positions and includes an explicitly solvable exponential-memory case [9]. Generalized Langevin and Mori–Zwanzig constructions provide the broader principle that memory equations can become local after the state is augmented [10–12].

The claims of this note are consequently structural and modest. The specific object is a discrete point process coupled to a nonlinear potential generated by a deposited memory field. For this object we make the normalized and finite-measure conventions explicit, prove the direct Markov and contraction statements, and specify what would count as metastability beyond a linear relaxation cloud. We do not prove ergodicity, a spectral gap, a nonlinear bifurcation, or the existence of a separate knot phase.

II. DISCRETE MODEL

Let $\Omega \subseteq \mathbb{R}^d$ be an effective state space with reference measure dx. The Euclidean structure is used only to write kernels, gradients, and diagnostics; it is not identified with physical space. Let G_σ be a smooth normalized deposition kernel of width σ, and let K be an interaction kernel for which the convolution

$$(K * \rho)(x) = \int_\Omega K(x, y)\rho(y)\, dy$$

is differentiable in the regimes considered below.

Assumption 1 (Baseline regularity). *Let $\Omega \subseteq \mathbb{R}^d$ be Borel and equip it with a boundary or extension convention that keeps x_{n+1} in Ω. Assume $G_\sigma \in L^1(\Omega)$, $G_\sigma \geq 0$, and $m_G = \int_\Omega G_\sigma < \infty$. The noise law is a Borel probability measure with finite second moment. For every finite memory measure reached by the recursion, $K(x, y)$ is measurable in (x, y), continuously differentiable in x, and $\nabla_x K(x, \cdot)$ is integrable against that measure; the resulting drift is Borel measurable. For the formal continuum and Hessian approximations we additionally require local Lipschitz and growth bounds on the drift and the corresponding second x-derivatives of K near the linearization region.*

The discrete process is

$$x_{n+1} = x_n + \varepsilon\xi_n - \eta\nabla\Phi_n(x_n), \tag{1}$$

$$\rho_{n+1}(x) = (1 - \lambda_\mathrm{m})\rho_n(x) + \beta G_\sigma(x - x_{n+1}), \tag{2}$$

$$\Phi_n(x) = (K * \rho_n)(x), \tag{3}$$

where $0 < \lambda_\mathrm{m} < 1$, $\beta \geq 0$, $\varepsilon \geq 0$, $\eta \geq 0$, and ξ_n are independent centered noise variables, typically taken isotropic with unit covariance. The parameter λ_m is the forgetting rate, while β is the deposited memory strength. The previously used normalized convention is the special case $\lambda_\mathrm{m} = \beta = \alpha$.

Let $m_G = \int_\Omega G_\sigma(x)\, dx$ and $M_n = \int_\Omega \rho_n(x)\, dx$. Then

$$M_{n+1} = (1 - \lambda_\mathrm{m})M_n + \beta m_G. \tag{4}$$

Thus the stationary total retained memory mass, when the scalar recursion is stable, is $M_* = \beta m_G/\lambda_\mathrm{m}$. Unit mass is preserved only in the normalized case $m_G = 1$, $M_0 = 1$, and

$\beta = \lambda_{\mathrm{m}}$. If ρ_0, G_σ, and β are nonnegative, every ρ_n is nonnegative. The choice $\beta = \lambda_{\mathrm{m}}$ is therefore a useful density normalization, not a structural requirement; an overall memory scale can often be absorbed into K or η.

The project term "knot" is reserved for a nonlinear metastable regime that survives comparison with the linear center-relative mode and matched controls. Compact motion around a moving memory center alone is called a localized memory cloud and does not meet that criterion.

III. EXPONENTIAL MEMORY AND CONTRACTIVE FIBRES

The memory equation can be unrolled exactly.

Proposition 1 (Exponential memory representation)**.** *For $n \geq 1$,*

$$\rho_n = (1-\lambda_{\mathrm{m}})^n \rho_0 + \beta \sum_{j=0}^{n-1} (1-\lambda_{\mathrm{m}})^j G_\sigma(\,\cdot - x_{n-j}). \tag{5}$$

Proof. The claim follows by induction. For $n = 1$ it is exactly Eq. (2). If it holds at n, then

$$\begin{aligned}\rho_{n+1} = {} & (1-\lambda_{\mathrm{m}})^{n+1}\rho_0 \\ & + \beta \sum_{j=1}^{n} (1-\lambda_{\mathrm{m}})^j G_\sigma(\cdot - x_{n+1-j}) \\ & + \beta G_\sigma(\cdot - x_{n+1}),\end{aligned}$$

which is Eq. (5) with $n+1$. □

The deposited weights are

$$w_j = \beta(1-\lambda_{\mathrm{m}})^j. \tag{6}$$

Their relative tail is controlled by λ_{m}, independently of the deposition scale β. For a rest-weight threshold δ, the effective memory depth satisfies

$$j_\delta = \frac{\log\delta}{\log(1-\lambda_{\mathrm{m}})} \sim \frac{|\log\delta|}{\lambda_{\mathrm{m}}} \quad (\lambda_{\mathrm{m}} \ll 1). \tag{7}$$

Thus $\lambda_{\mathrm{m}}^{-1}$ is the natural update-count scale for memory persistence. Figure 1 shows the normalized convention $\beta = \lambda_{\mathrm{m}} = \alpha$ for representative values of α.

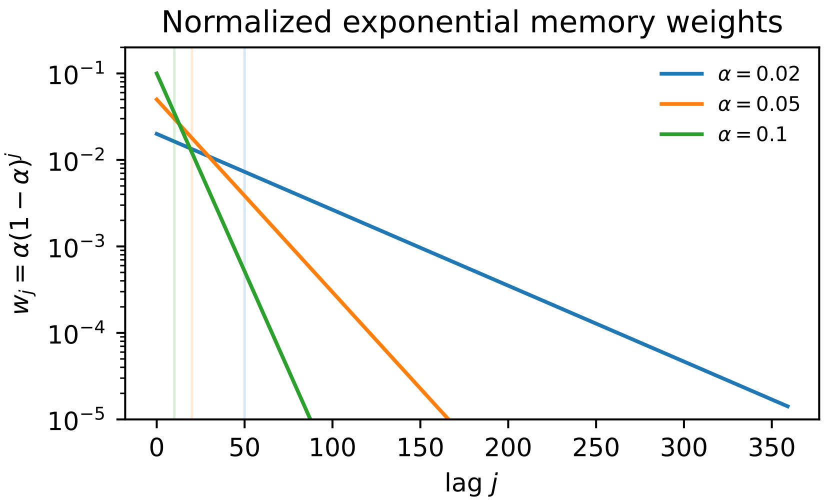

FIG. 1. Normalized exponential memory weights $w_j = \alpha(1-\alpha)^j$, corresponding to $\beta = \lambda_{\mathrm{m}} = \alpha$. Vertical guide lines mark $j = \lambda_{\mathrm{m}}^{-1} = \alpha^{-1}$ for each curve.

Proposition 2 (Pathwise memory-fibre contraction)**.** *Fix a common visible trajectory $x_1, \ldots, x_n$. Let ρ_k and $\tilde{\rho}_k$ solve Eq.* (2) *with the same deposited kernels but different initial memory fields. Then*

$$\rho_n - \tilde{\rho}_n = (1-\lambda_{\mathrm{m}})^n(\rho_0 - \tilde{\rho}_0). \tag{8}$$

Consequently, for any homogeneous norm on the memory space,

$$\|\rho_n - \tilde{\rho}_n\| = (1-\lambda_{\mathrm{m}})^n\|\rho_0 - \tilde{\rho}_0\|. \tag{9}$$

Proof. Subtract the two memory recursions. The deposited kernels cancel because the visible trajectory is common:

$$\rho_{k+1} - \tilde{\rho}_{k+1} = (1-\lambda_{\mathrm{m}})(\rho_k - \tilde{\rho}_k).$$

Iterating gives Eq. (8). □

This contraction is a statement about the memory fibre under a fixed visible path. It is not yet a global contraction theorem for the full feedback system, because different memory fields can alter future positions through Eq. (1). A full contraction analysis must control the coupled pair (x_n, ρ_n).

IV. MARKOVIAN EMBEDDING

Define the augmented state

$$z_n = (x_n, \rho_n) \in \Sigma. \tag{10}$$

Given z_n and a fresh noise draw ξ_n, Eqs. (1) and (2) determine z_{n+1}. Thus the augmented dynamics is a time-homogeneous Markov process.

Definition 1 (Transition kernel)**.** *For measurable $A \subseteq \Sigma$ define*

$$P(z, A) = \mathbb{P}\{z_{n+1} \in A \mid z_n = z\}. \tag{11}$$

The observable Markov operator is

$$(Uf)(z) = \int_\Sigma f(z')P(z, dz') = \mathbb{E}[f(z_{n+1}) \mid z_n = z]. \tag{12}$$

The operator U is linear, positive, and unital: $U1 = 1$. Its iterates satisfy $U^{m+n} = U^m U^n$. In the stochastic case it is generally not multiplicative: $U(fg)$ need not equal $(Uf)(Ug)$. The correct algebraic object is therefore a positive unital Markov operator on observables, with a dual action on probability measures, rather than a deterministic algebra automorphism.

Proposition 3 (Visible non-Markovian projection)**.** *The projected process x_n is generally not Markov.*

Proof. Take the same visible position x with two memory fields ρ and $\tilde{\rho}$ such that

$$\nabla(K * \rho)(x) \neq \nabla(K * \tilde{\rho})(x).$$

Then the conditional next-step laws of x_{n+1} differ because their drift terms in Eq. (1) differ. Hence conditioning on $x_n = x$ alone does not determine the next-step law. □

The memory update should not be described as algebraically noninvertible in isolation. If x_{n+1} and ρ_{n+1} are known and $\lambda_\mathrm{m} \neq 1$, then

$$\rho_n = \frac{\rho_{n+1} - \beta G_\sigma(\cdot - x_{n+1})}{1 - \lambda_\mathrm{m}}$$

is a formal rearrangement. The precise irreversibility statement is different: the present augmented state does not generally encode the full ordered history, and the stochastic dynamics defines a forward Markov semigroup rather than a reversible group action.

V. FORMAL CONTINUUM APPROXIMATION

Let $t = \lambda_{\mathrm{m}} n$ and consider a joint scaling

$$\lambda_{\mathrm{m}} \to 0, \qquad \varepsilon \to 0, \qquad \frac{\varepsilon^2}{2\lambda_{\mathrm{m}}} \to D,$$
$$\frac{\eta}{\lambda_{\mathrm{m}}} \to \gamma, \qquad \frac{\beta}{\lambda_{\mathrm{m}}} \to \kappa.$$

The final scaling assumption keeps the memory source finite; the normalized case has $\kappa = 1$. Formally, the visible coordinate approaches

$$dX_t = -\gamma \nabla (K * \rho_t)(X_t)\, dt + \sqrt{2D}\, dW_t, \tag{13}$$

while the memory field satisfies

$$\partial_t \rho_t = \kappa G_\sigma(\cdot - X_t) - \rho_t. \tag{14}$$

More generally one may introduce a memory time τ and write

$$\partial_t \rho_t = \tau^{-1} \left(\kappa G_\sigma(\cdot - X_t) - \rho_t \right),$$

equivalently

$$\rho_t = \kappa \tau^{-1} \int_{-\infty}^{t} e^{-(t-s)/\tau} G_\sigma(\cdot - X_s)\, ds$$

after transients. The extended process (X_t, ρ_t) is local in the scaled time variable and Markovian; the visible process X_t alone retains memory.

Equations (13)–(14) are a formal effective description, not a convergence theorem in this paper. A rigorous limit would require specifying topology, tightness estimates, regularity of K, boundary conditions, and the scaling of Ω if the domain changes with λ_{m}.

VI. METASTABILITY AND KNOT DIAGNOSTICS

A candidate knot is a long-lived localized feedback regime in the augmented dynamics. Operationally, for a region $\mathcal{K} \subseteq \Omega$ one can track retained memory mass

$$M_n(\mathcal{K}) = \int_{\mathcal{K}} \rho_n(x)\, dx. \tag{15}$$

Persistence over many memory times,

$$\Delta n \gg \lambda_{\mathrm{m}}^{-1},$$

is a minimal diagnostic for metastability, not a proof of a new particle-like object.

Several diagnostics are useful and mutually constraining:

- residence-time statistics of x_n in candidate regions;
- retained memory weight $M_n(\mathcal{K})$;
- autocorrelation functions of positions and reduced memory features;
- local linearization near a metastable center x_*;
- slow modes of a transfer-operator estimate on augmented features.

For local linearization, suppose a metastable regime has slowly varying memory ρ and a center x_* with $\nabla(K * \rho)(x_*) = 0$. With

$$H = \nabla\nabla(K * \rho)(x_*)$$

the continuum approximation becomes locally

$$dX_t = -\gamma H(X_t - x_*)\, dt + \sqrt{2D}\, dW_t. \tag{16}$$

Positive eigenvalues of γH give local relaxation rates in stable directions. A relaxation-based stability scale can be written as

$$\Gamma_{\text{rel}} = \tau_{\text{rel}}^{-1}, \qquad \widehat{\Gamma} = \frac{\sigma^2}{D}\Gamma_{\text{rel}}.$$

This is a mass-like or stability proxy only. It is not a calibrated physical mass and should not be identified with mc^2.

A. Transfer-Operator View

Let $\psi(z_n)$ be a reduced feature vector for the augmented state, for example position, memory centroid, current offset from the memory centroid, memory spread, retained memory mass, or learned/PCA/Fourier features of ρ_n. A finite-rank Ulam or Markov-state approximation estimates a transition matrix at sample lag ℓ from pairs

$$(\psi(z_n), \psi(z_{n+\ell})).$$

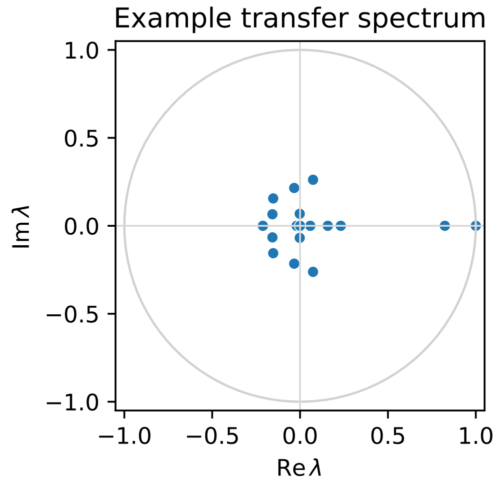

FIG. 2. Example eigenvalues of a finite-state transfer estimate from reduced augmented features. Non-unit modes close to the unit circle are candidates for metastable structure, subject to lag-time choice, discretization, finite-sample row handling, and sampling checks.

Let τ_ℓ denote the update-time or scaled-time lag represented by those pairs. For finite trajectories, states that are observed only as terminal lagged targets have no empirical outgoing row. The reference implementation reports how many such rows are handled and, by default, treats them as absorbing in the finite matrix. Eigenvalues with modulus numerically equal to one are still reported in the spectrum but are not converted into relaxation rates. Production analyses should check the terminal-row count, vary the lag and discretization, and compare against independent residence diagnostics. Slow non-unit eigenvalues $\lambda_i(\tau_\ell)$ then give implied rates

$$\Gamma_i(\tau_\ell) = -\frac{1}{\tau_\ell} \log |\lambda_i(\tau_\ell)|. \tag{17}$$

Almost-invariant sets and coherent sets are then candidate metastable knots [13]. Related variational and Koopman approaches are standard in molecular kinetics and operator-theoretic model reduction [14–17].

Figure 2 shows one small reproducible transfer spectrum from the companion pipeline. It is an illustration of the diagnostic workflow, not evidence for a universal phase diagram.

VII. RELATION TO EXISTING LITERATURE

The normalized occupation measure of Benaïm, Ledoux, and Raimond weights the past cumulatively as $t^{-1}\int_0^t \delta_{X_s}\,ds$. Their convergence and free-energy results, including the symmetric-interaction theory of Benaïm and Raimond, therefore concern a different time dependence from Eq. (2) [6, 7]. Herrmann and Roynette use the unnormalized full-history drift $\int_0^t \Phi(X_t - X_s)\,ds$ and prove strong localization statements for specific one-dimensional interactions [8]. This is an important nearby localization result, but it is not an exponential-memory model.

Milišić, Meunier, and Roux introduce an aging kernel for linear self-interaction and analyze general convolution weights as well as a particular exponential case [9]. This is the closest temporal-memory comparison found in the present review. The model here differs by using a deposited spatial density and allowing a nonlinear kernel-induced force, but the general ideas of aging and exponential state closure are shared. Accordingly, novelty cannot rest on exponential forgetting or Markov augmentation alone.

Mori–Zwanzig and generalized Langevin methods provide the broader embedding principle [10–12]. Transfer operators and Markov-state models provide standard diagnostics for slow processes and almost-invariant sets [14–17]. The value of the present note is the explicit specification and separation of these ingredients for the companion scalar model, not a new general embedding theorem.

VIII. OPEN MATHEMATICAL PROBLEMS

The structural results above do not settle the main hard questions. The following are open tasks:

- well-posedness of the continuum approximation under explicit kernel and state-space assumptions;
- Feller properties and invariant measures for the augmented Markov process;
- Lyapunov/Harris drift conditions of the form $PV \leq \lambda V + b$ outside compact sets [18];
- conditions for uniqueness, geometric ergodicity, and spectral gaps;

- coupled contraction estimates for (x_n, ρ_n), not only the memory fibre under a fixed path;
- bifurcation and metastability regimes for attractive, repulsive, and two-scale attractive-repulsive kernels;
- convergence and bias analysis for transfer-operator estimates on reduced memory features.

IX. BOUNDARIES

The structural results do not imply a nonlinear knot phase, dimension selection, a hard finite signal speed, Lorentz invariance, quantum dynamics, particle structure, or calibrated masses. A finite propagation bound would additionally require local transfer and the absence of direct long-range response. Dimension-selection claims require ambient-dimension controls and a diagnostic that does not simply reproduce isotropic covariance rank. Relaxation rates remain model diagnostics unless an independent physical calibration is supplied.

X. CONCLUSION

The model defines a self-interacting stochastic process with exponential forgetting and a field-mediated self-potential. The visible trajectory is generally non-Markovian, but the explicit memory field yields a Markovian augmented dynamics. The memory fibre is affine and pathwise contractive for a common visible trajectory. These structural facts make localization and metastability testable, but they do not prove a nonlinear metastable regime. The note supplies the mathematical reference needed to formulate and falsify such a claim in the companion numerical study.

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[1] R. Pemantle, Probability Surveys **4**, 1 (2007).

[2] R. T. Durrett and L. C. G. Rogers, Probability Theory and Related Fields **92**, 337 (1992).

[3] D. J. Amit, G. Parisi, and L. Peliti, Physical Review B **27**, 1635 (1983).

[4] B. Tóth and W. Werner, Probability Theory and Related Fields **111**, 375 (1998).

[5] L. Peliti and L. Pietronero, La Rivista del Nuovo Cimento **10**, 1 (1987).

[6] M. Benaïm, M. Ledoux, and O. Raimond, Probability Theory and Related Fields **122**, 1 (2002).

[7] M. Benaïm and O. Raimond, The Annals of Probability **33**, 1716 (2005).

[8] S. Herrmann and B. Roynette, Mathematische Annalen **325**, 81 (2003).

[9] V. Milišić, N. Meunier, and R. Roux, Self-interacting diffusions with aging, SSRN preprint (2026).

[10] H. Mori, Progress of Theoretical Physics **33**, 423 (1965).

[11] R. Zwanzig, *Nonequilibrium Statistical Mechanics* (Oxford University Press, 2001).

[12] D. Jaganathan and R. N. Valani, Communications in Nonlinear Science and Numerical Simulation 10.1016/j.cnsns.2025.109540 (2025).

[13] G. Froyland, S. Lloyd, and N. Santitissadeekorn, Physica D **239**, 1527 (2010).

[14] F. Noé and F. Nüske, Multiscale Modeling & Simulation **11**, 635 (2013).

[15] F. Noé, H. Wu, J.-H. Prinz, and N. Plattner, The Journal of Chemical Physics **139**, 184114 (2013).

[16] M. O. Williams, I. G. Kevrekidis, and C. W. Rowley, Journal of Nonlinear Science **25**, 1307 (2015).

[17] S. Klus, F. Nüske, P. Koltai, H. Wu, I. Kevrekidis, C. Schütte, and F. Noé, Journal of Nonlinear Science **28**, 985 (2018).

[18] S. Meyn and R. L. Tweedie, *Markov Chains and Stochastic Stability*, 2nd ed. (Cambridge University Press, 2009).